Base case: for $n=1$ $\quad 1=1^2$

Induction hypothesis: $1+3+\ldots+(2k-1) = k^2$

Induction step: $1+3+\ldots+(2k-1)+2k+1=(k+1)^2$

$\Rightarrow k^2+2k+1 = k^2+2k+1$